Program Name	**B.C.A**	Semester	**VI**
Course Title	**Fundamentals of Data Science (Theory)**		
Course Code:	**DSE-E2**	No. of Credits	**03**
Contact hours	**42 Hours**	Duration of SEA/Exam	**2 1/2 Hours**
Formative Assessment Marks	**40**	Summative Assessment Marks	**60**

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1	Understand the concepts of data and pre-processing of data.
CO2	Know simple pattern recognition methods
CO3	Understand the basic concepts of Clustering and Classification
CO4	Know the recent trends in Data Science

Contents	**42 Hrs**
Unit I: Data Mining: Introduction, Data Mining Definitions, Knowledge Discovery in Databases (KDD) Vs Data Mining, DBMS Vs Data Mining, DM techniques, Problems,Issues and Challenges in DM, DM applications.	**8**
Data Warehouse: Introduction, Definition, Multidimensional Data Model, Data Cleaning, Data Integration and transformation, Data reduction, Discretization	**8**
Mining Frequent Patterns: Basic Concept – Frequent Item Set Mining Methods - Aprioriand Frequent Pattern Growth (FPGrowth) algorithms -Mining Association Rules	**8**
Classification: Basic Concepts, Issues, Algorithms: Decision Tree Induction. Bayes Classification Methods, Rule-Based Classification, Lazy Learners (or Learning from yourNeighbors), k Nearest Neighbor. Prediction - Accuracy- Precision and Recall	**10**
Clustering: Cluster Analysis, Partitioning Methods, Hierarchical Methods, Density-Based Methods, Grid-Based Methods, Evaluation of Clustering	**8**

Unit 2

Topics:

Data Warehouse: Introduction, Definition, Multidimensional Data Model, Data Cleaning, Data Integration and transformation, Data reduction, Discretization.

Data Warehouse:

Def 1: A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process."—W. H. Inmon (Father of Data Warehouse- American Computer Scientist)

Def 2: Centralized data location for multiple sources of data.

Data warehousing provides architectures and tools for business executives to systematically organize, understand, and use their data to make strategic decisions.
According to William H. Inmon, a leading architect in the construction of data warehouse systems, "A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision making process"
Subject-oriented: A data warehouse is organized around major subjects such as customer, supplier, product, and sales.
Integrated: A data warehouse is usually constructed by integrating multiple heterogeneous sources, such as relational databases, flat files, and online transaction records.
Time-variant : Data are stored to provide information from an historic perspective.
(e.g., the past 5–10 years).
Nonvolatile: A data warehouse is always a physically separate store of data transformed from the application data found in the operational environment. Due to this separation, a data warehouse does not require transaction processing, recovery, and concurrency control mechanisms. It usually requires only two operations in data accessing: initial loading of data and access of data

Data warehousing:

The process of constructing and using data warehouses as shown the following figure.

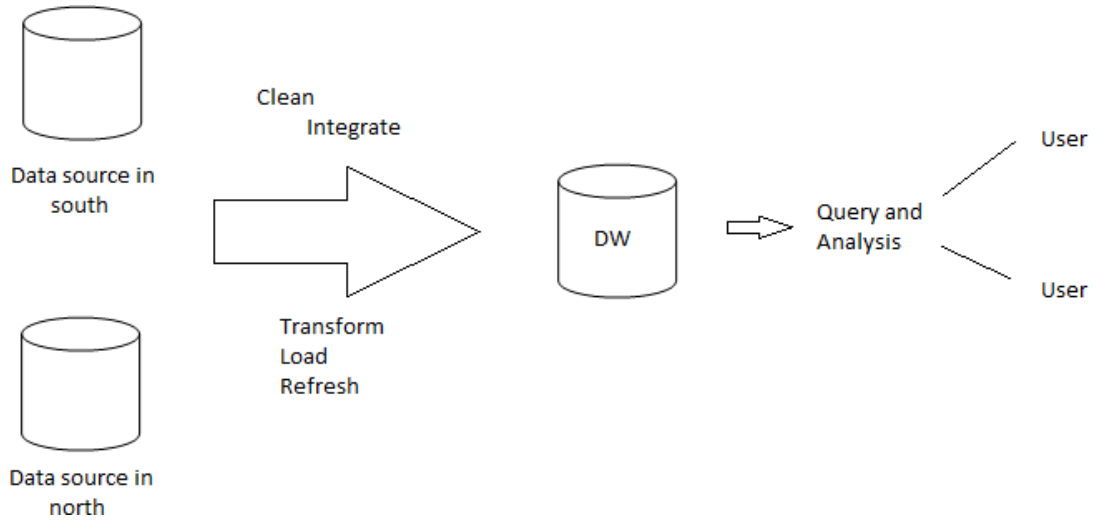

Fig 1.1: Data ware house of a sales organization.

Difference between

	OLTP	**OLAP**
users	clerk, IT professional	knowledge worker
function	day to day operations	decision support
DB design	application-oriented	subject-oriented
data	current, up-to-date detailed, flat relational isolated	historical, summarized, multidimensional integrated, consolidated
usage	repetitive	ad-hoc
access	read/write index/hash on prim. key	lots of scans
unit of work	short, simple transaction	complex query
# records accessed	tens	millions

#users	thousands	hundreds
DB size	100MB-GB	100GB-TB
metric	transaction throughput	query throughput, response

Data Warehousing: Three Tier Architecture

Data warehouses often adopt a three-tier architecture, as presented in Figure.

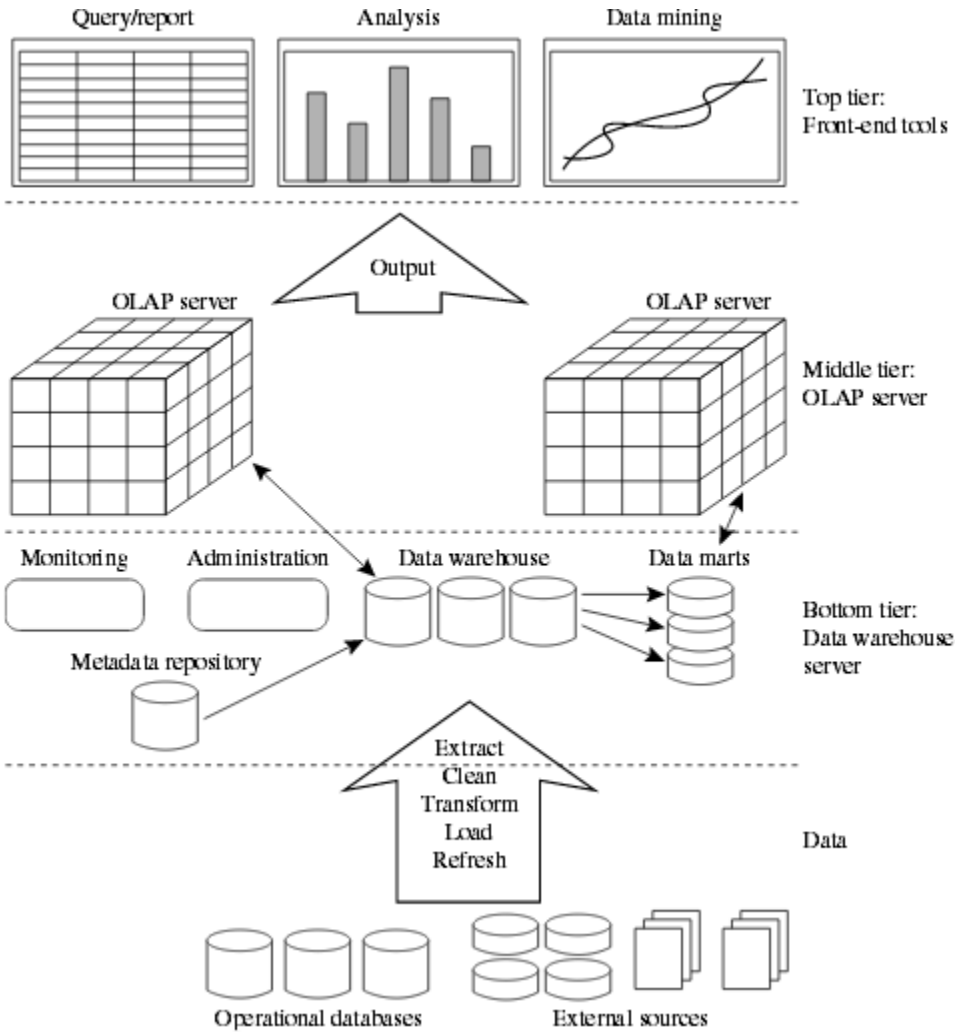

Fig. Three Tier Architecture of Data warehousing

- The bottom tier is a warehouse database server that is almost always a relational database system. Back-end tools and utilities are used to feed data into the bottom tier from operational databases or other external sources (e.g., customer profile information provided by external consultants). These tools and utilities perform data extraction, cleaning, and transformation (e.g., to merge similar data from different sources into a unified format), as well as load and refresh functions to update the data warehouse

- The middle tier is an OLAP server that is typically implemented using either

  (1) a relational OLAP (ROLAP) model (i.e., an extended relational DBMS that maps operations on multidimensional data to standard relational operations); or

  (2) a Multi-dimensional OLAP (MOLAP) model (i.e., a special-purpose server that directly implements multidimensional data and operations).

- The top tier is a front-end client layer , which contains query and reporting tools, analysis tools, and/or data mining tools (e.g., trend analysis, prediction, and so on).

Data Warehouse Models: Enterprise Warehouse, Data Mart, and Virtual Warehouse

- Enterprise warehouse
  - collects all of the information about subjects spanning the entire organization
- Data Mart
  - a subset of corporate-wide data that is of value to a specific groups of users. Its scope is confined to specific, selected groups, such as marketing data mart
- Virtual warehouse
  - A set of views over operational databases
  - Only some of the possible summary views may be materialized

A recommended method for the development of data warehouse systems is to implement the warehouse in an incremental and evolutionary manner, as shown in Figure.

First, a high-level corporate data model is defined within a reasonably short period (such as one or two months) that provides a corporate-wide, consistent, integrated view of data among different subjects and potential usages. This high-level model, although it will need to be refined in the further development of enterprise data warehouses and departmental data marts, will greatly reduce future integration problems. Second, independent data marts can be implemented in parallel with the enterprise warehouse based on the same corporate data model set noted before. Third, distributed data marts can be constructed to integrate different data marts via hub servers. Finally, a multitier data warehouse is constructed where the enterprise warehouse is the sole custodian of all warehouse data, which is then distributed to the various dependent data marts.

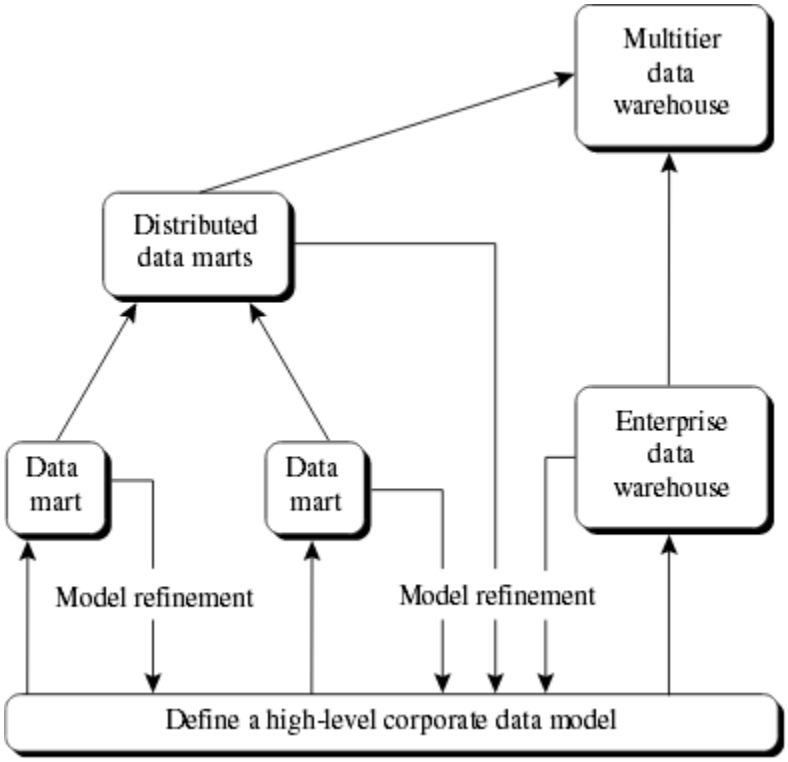

Fig: A recommended approach for data warehouse development

Data Warehouse Modeling: Data Cube and OLAP

Data warehouses and OLAP tools are based on a multidimensional data model. This model views data in the form of a data cube.

- A data cube, such as sales, allows data to be modeled and viewed in multiple dimensions. It is defined by dimensions and facts. Fact tables contain numerical data, while dimension tables provide context and background information.
  - **Dimension tables**, such as item (item_name, brand, type), or time(day, week, month, quarter, year) (entities in which org keeps records)
  - **Fact table** contains numeric **measures** (such as dollars_sold(sale amt in $), units sold) and keys to each of the related dimension tables

In data warehousing literature, an n-D base cube is called a base cuboid. The top most 0-D cuboid, which holds the highest-level of summarization, is called the apex cuboid. The apex cuboid is typically denoted by ‘all’.

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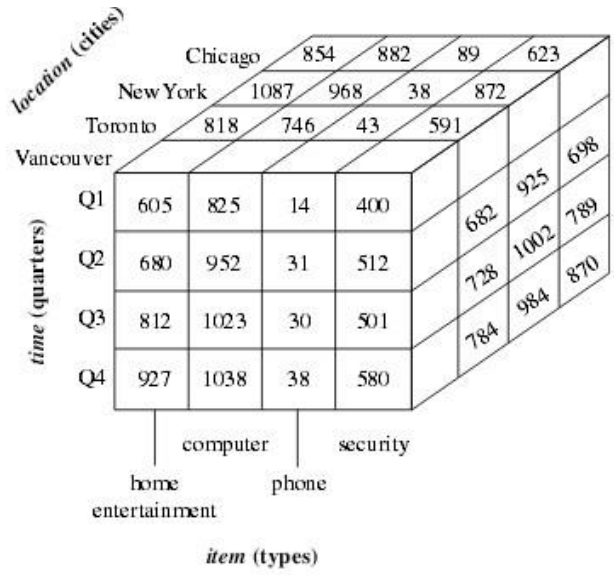


Figure 4.3 A 3-D data cube representation of the data in Table 4.3, according to *time*, *item*, and *location*. The measure displayed is *dollars_sold* (in thousands).

The lattice(patterened structure like fence) of cuboids forms a data cube as shown below.

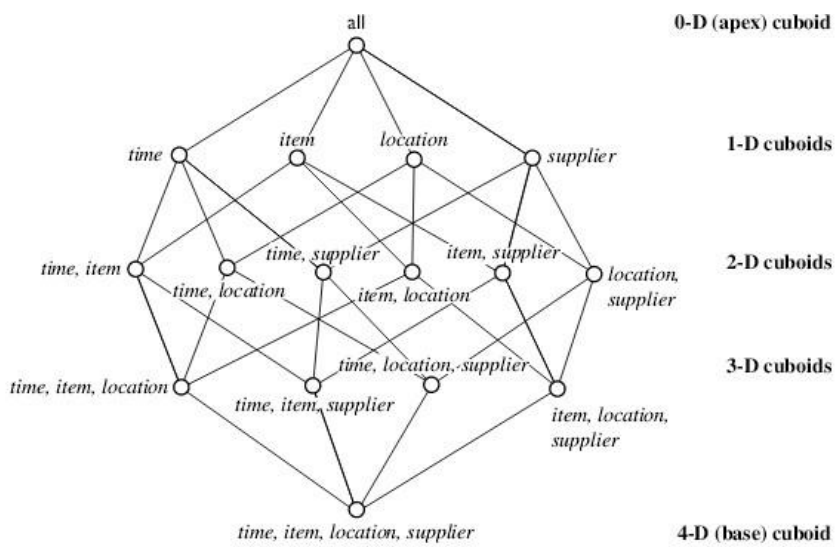


Figure 4.5 Lattice of cuboids, making up a 4-D data cube for *time*, *item*, *location*, and *supplier*. Each cuboid represents a different degree of summarization.

Schemas for Multidimensional Data Models

Stars, Snowflakes, and Fact Constellations:

- <u>Star schema</u>: A fact table in the middle connected to a set of dimension tables

- Snowflake schema: A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake

- Fact constellations: Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation

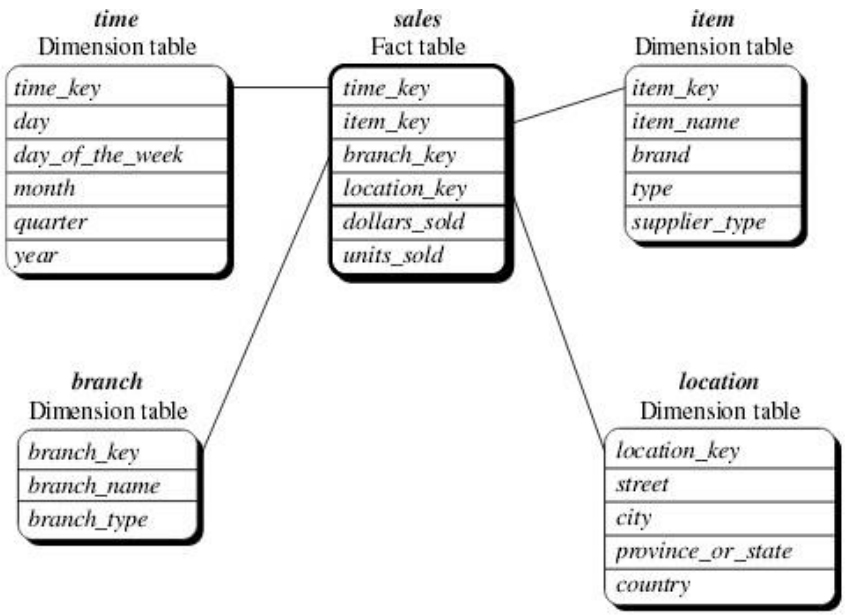


Figure 4.6 Star schema of *sales* data warehouse.

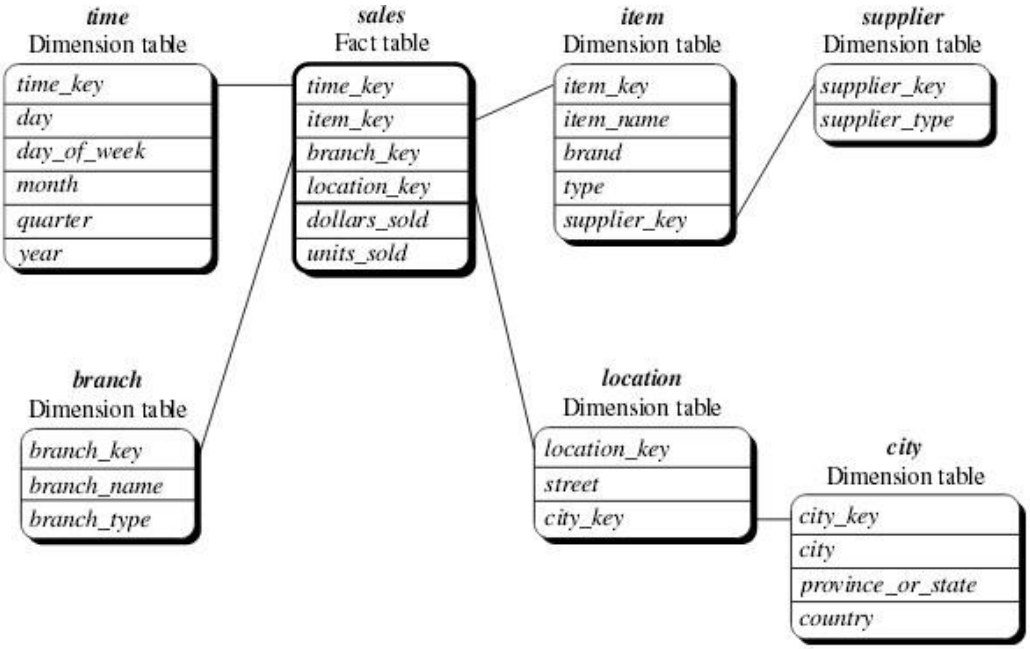


Figure 4.7 Snowflake schema of a *sales* data warehouse.

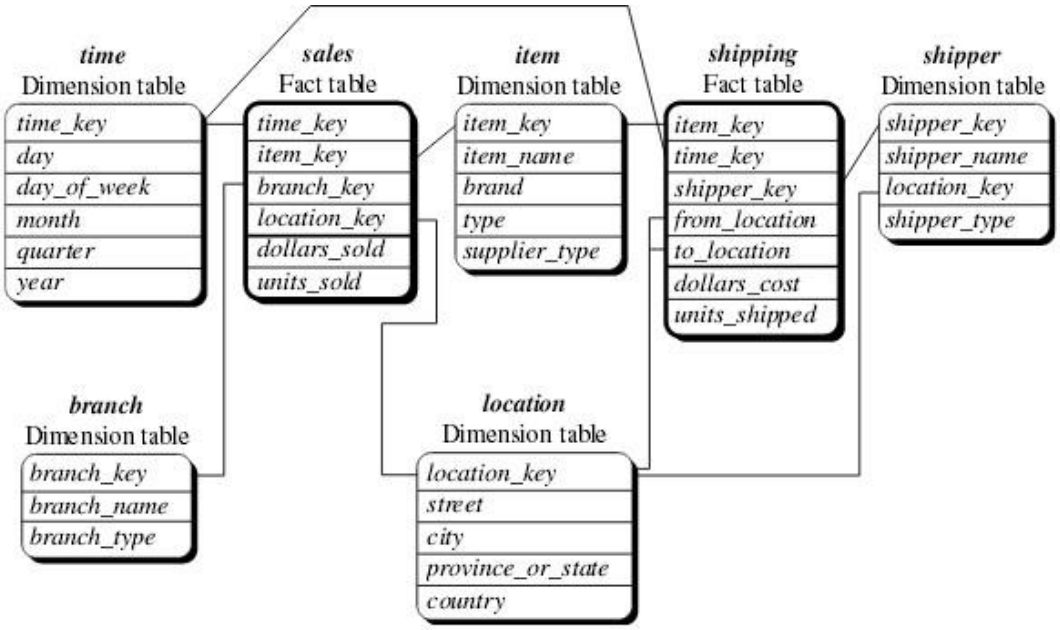
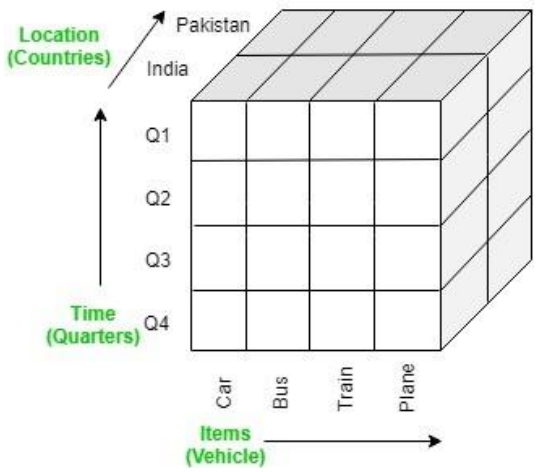


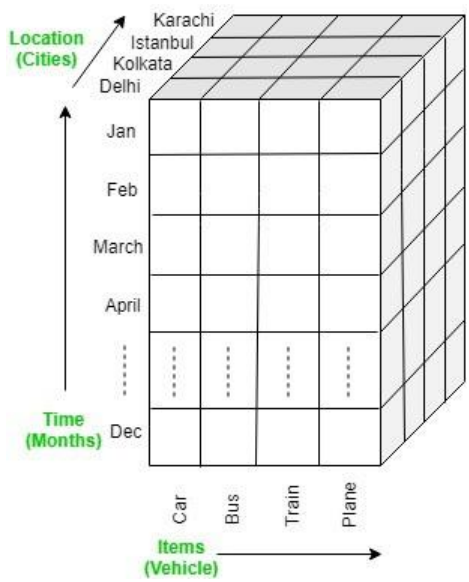
Figure 4.8 Fact constellation schema of a sales and shipping data warehouse.

OLAP Operations

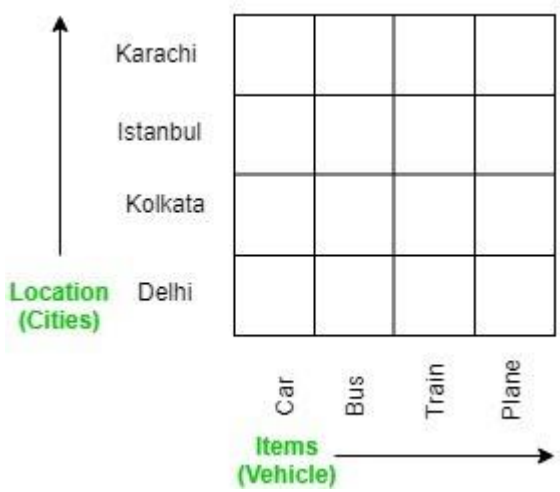
- Roll up (drill-up): summarize data or aggregation of data
  - *by climbing up hierarchy or by dimension reduction*
  - In the cube given in the overview section, the roll-up operation is performed by climbing up in the concept hierarchy of *Location* dimension (City -> Country).



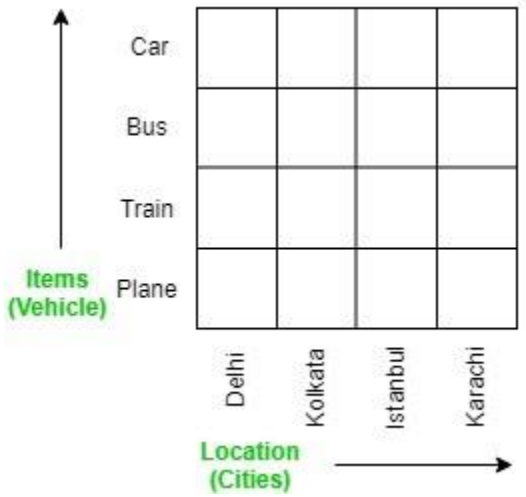
- Drill down (roll down): *In drill-down operation, the less detailed data is converted into highly detailed data. It can be done by:*
  - *Moving down in the concept hierarchy*
  - *Adding a new dimension*
  - In the cube given in overview section, the drill down operation is performed by moving down in the concept hierarchy of *Time* dimension (Quarter -> Month).



- Slice and dice: It selects a single dimension from the OLAP cube which results in a new sub-cube creation. In the cube given in the overview section, Slice is performed on the dimension Time = "Q1"



- Pivot (rotate):
  - *reorient the cube, visualization, 3D to series of 2D planes*
  - It is also known as *rotation* operation as it rotates the current view to get a new view of the representation. In the sub-cube obtained after the slice operation, performing pivot operation gives a new view of it.



Data Cleaning, …

Today's data are highly susceptible to noisy, missing and inconsistent data due to their typically huge size and because of heterogeneous sources. Low quality data will lead to poor mining results.

Different data preprocessing techniques(data cleaning, data integration, data reduction, data transformation) that when applied before data mining will improve the overall quality of the pattern mined and also time required for actual mining.

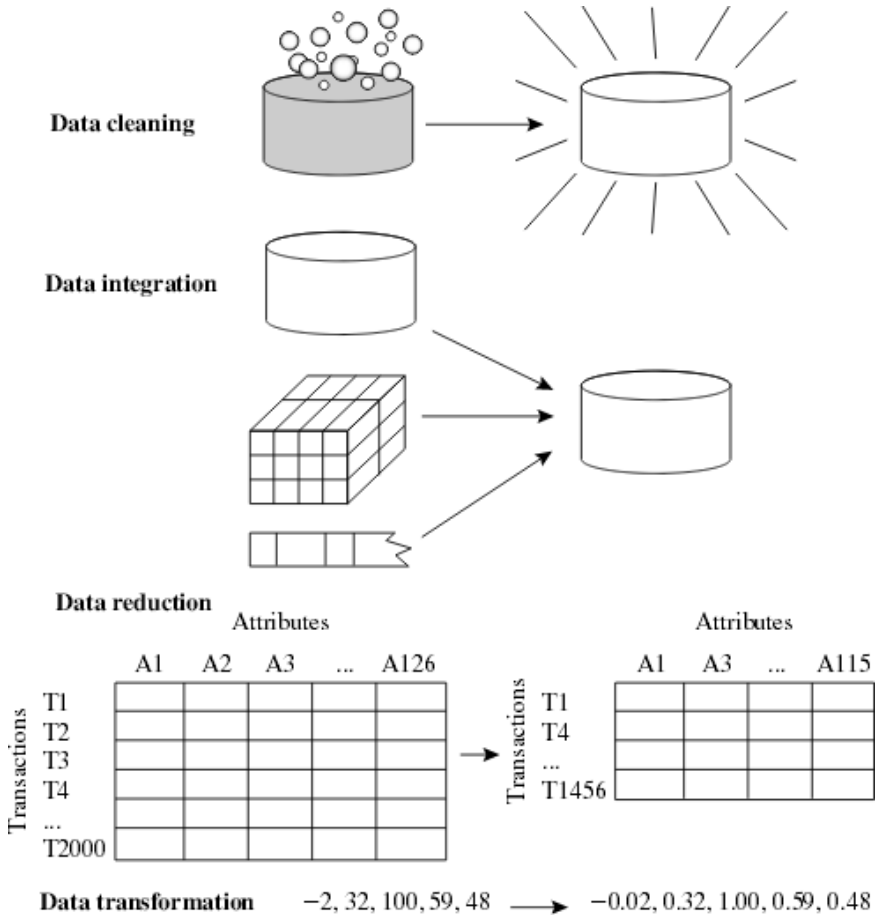


Fig: Forms of Data Preprocessing

<u>Data cleaning</u>

Data cleaning stage helps in smooth out noise, attempts to fill in missing values, removing outliers, and correct inconsistency in data.

1) **Handling missing values:**

i. **Ignoring the tuple**: Used when class label is missing. This method is not very effective when more missing value is present.
ii. **Fill in missing value manually**: It is time consuming.
iii. **Using global constant to fill missing value**: Ex: unknown or ∞
iv. **Use attribute mean to fill the missing value**
v. **Use attribute mean for all samples belonging to the same class as the given tuple**
vi. **Use most probable value to fill the missing value**: (using decision tree)

2) Noisy data: Noise is a random error or variance in measured variable.

Different methods for smoothing are:

1. **Binning:** Smooth the sorted data by consulting its neighborhood. The values are distributed into buckets/bins. They perform local smoothing.

Different binning methods for data smoothing:

i. **Smoothing by bin means:** Each value in bin is replaced by mean
Ex: BIN 1 : 4,8,15 = BIN 1: 9,9,9
ii. **Smoothing by bin boundaries:** Min and max value is identified and value is replaced by closest boundary value
Ex: BIN 1 : 4,8,15 = BIN 1: 4,4,15

2. **Regression:** Data smoothing can also be done by regression(linear regression, multiple linear regression). In this one attribute can be used to predict the value of another.
3. **Outlier analysis:** Outliers can be done by clustering. The value outside the clusters are outliers.

<u>Data Integration</u>

Data mining often works on integrated data from multiple repositories. Careful integration helps in accuracy of data mining results.

<u>Challenges of DI</u>

1. **Entity Identification Problem:**
"How to match schema and objects from many sources?" This is called Entity Identification Problem.
Ex: Cust-id in one table and Cust-no in another table.
Metadata helps in avoiding these problems.
2. **Redundancy and correlation analysis:**
Redundancy -> repetition.
Some redundancy can be detected by correlation analysis. Given two attributes, correlation tell how strongly the relationship is (Chi-square test, correlation coefficient are ex).

<u>Data Reduction</u>

Data Reduction techniques can be applied to obtain a reduced representation of the data set that is much smaller in volume, yet closely maintain the integrity of the original data.

Data Reduction Strategies:

1. **Dimensionality reduction:**
Reducing the number of attributes/variables under consideration.

Ex: Attribute subset selection, Wavelet Transform, PCA.

2. **Numerosity reduction:**
   Replace original data by alternate smaller forms.
   Ex: Histograms, Sampling, Data cube aggregation.
3. **Data compression:**
   Reduce the size of data.

Wavelet Transform:

DWT- Discrete Wavelet Transform is a linear signal processing technique, that when applied to a data vector X, transforms it to a numerically different vector X' of same length. The DWT is a fast and simple transformation that can translate an image from the spatial domain to the frequency domain.

Principal Components Analysis(PCA)

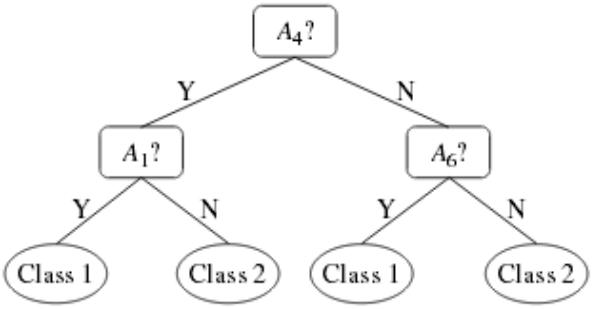
PCA reduces the number of variables or features in a data set while still preserving the most important information like major trends or patterns.

Attribute Subset Selection:

Dataset for analysis consists of many attribute which may be irrelevant to the mining task. (Ex: Telephone no. may not be important while classifying customer). Attribute subset selection reduces the data set by removing irrelevant attributes.

Some heuristics methods for attribute subset selection are:

1. Stepwise forward selection:
   - Start with empty set
   - Best attribute are added to reduce set
   - At each iteration, the rest of remaining attribute are added.
2. Stepwise backward elimination:
   - Start with full set of attributes
   - At each step, remove the worst attributes.
3. Combination of forward selection & backward selection:
   - Combined method
   - At each step, procedure selects the best attribute & remove worst from remaining.
4. Decision Tree Induction:
   In DTI a tree is constructed from the given data. All attributes that do not appear in tree are assumed to be irrelevant.

Forward selection	Backward elimination	Decision tree induction
Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ Initial reduced set: $\{\}$ => $\{A_1\}$ => $\{A_1, A_4\}$ => Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ => $\{A_1, A_3, A_4, A_5, A_6\}$ => $\{A_1, A_4, A_5, A_6\}$ => Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$  => Reduced attribute set: $\{A_1, A_4, A_6\}$

Histograms:

Histogram is a frequency plot. It uses bins/buckets to approximate data distributions and are popular form of data reduction. They are highly effective at approximating both sparse & dense data as well as skewed & uniform data.

The following data are a list of AllElectronics prices for commonly sold items (rounded to the nearest dollar). The numbers have been sorted: 1, 1, 5, 5, 5,5, 5, 8, 8, 10, 10, 10, 10, 12, 14, 14, 14, 15, 15, 15, 15, 15, 15, 18, 18, 18, 18, 18,18, 18, 18, 20, 20, 20, 20, 20, 20, 20, 21, 21, 21, 21, 25, 25, 25, 25, 25, 28, 28, 30,30, 30) Figure shows the histogram for this data.

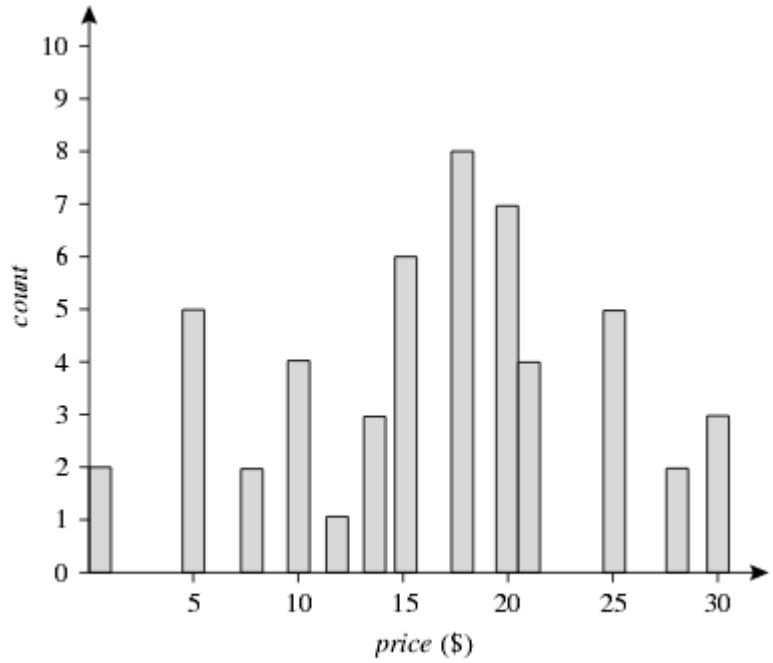


Fig : Histogram for ALL Electronics

Clustering:

Clustering partition data into clusters/groups which are similar/close. In data reduction, cluster representation of data are used to replace the actual data.

Sampling:

Used as data reduction technique in which large data are represented as small random samples (subset).

Common ways to sample:

i. Simple random sample without replacements of size(SRSWOR)

This is created by drawing *s* of the N tuples from D (s < N), where the probability of drawing any tuple in D is 1 = N , that is, all tuples are equally likely to be sampled.

ii. Simple random sample with replacement(SRSWR)

This is similar to SRSWOR, except that each time a tuple is drawn from , it is recorded and then replaced . That is, after a tuple is drawn, it is placed back in D so that it may be drawn again.

iii. Cluster sample

The tuples in D are grouped into M mutually disjoint "clusters," then an SRS of s clusters can be obtained, where s < M .

iv. Stratified sample

If D is divided into mutually disjoint parts called strata, a stratified sample of D is generated by obtaining an SRS at each stratum. For example, a stratified sample may be obtained from customer data, where a stratum is created for each customer age group. In this way, the age group having the smallest number of customers will be sure to be represented

An advantage of sampling for data reduction is that the cost of obtaining a sample is proportional to the size of the sample, s , as opposed to N , the data set size. Hence, sampling complexity is potentially sublinear to the size of the data.

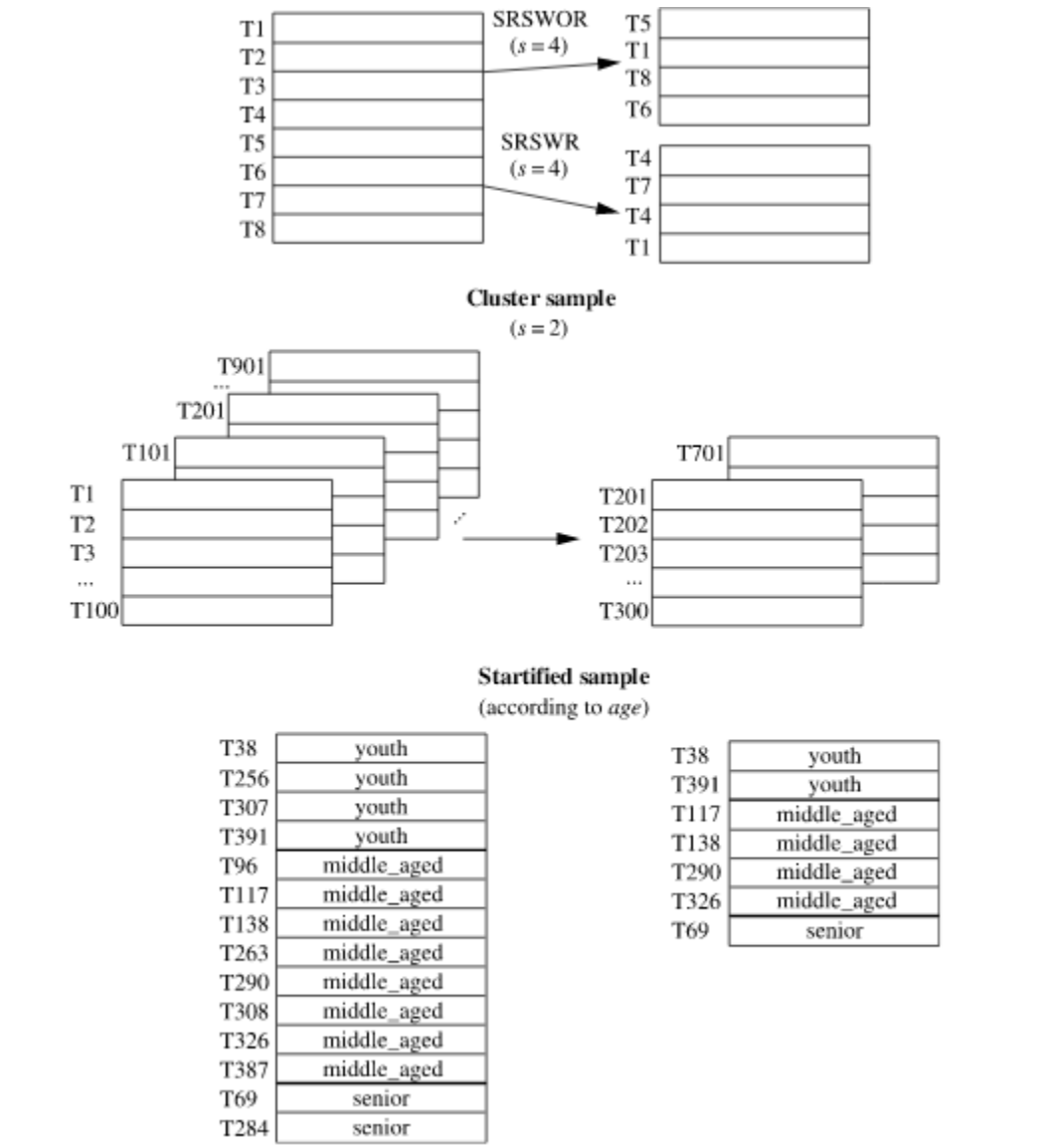


Fig. Sampling Techniques

Data Cube Aggregation:

- Aggregate data into one view.
- Data cube store multidimensional aggregated information.
- Data cube provides fast access to precomputed, summarized data, thereby benefits OLAP/DM.
- Data cube created for varying level of abstraction are often referred to as **cuboids**.
- Cube created at lowest level of abstraction is **base cuboids.**
    - Ex: Data regarding sales or customers.
- Cube created at highest level of abstraction is **apex cuboids.**
    - Ex: Total sales for all 3 years, for items.

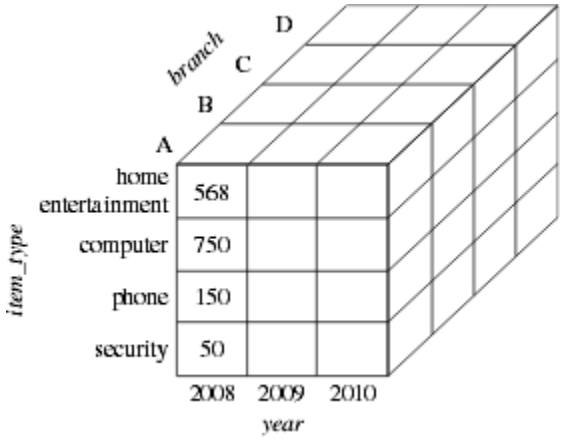


Fig. Data Cube

Data Transformation

The data is transformed or consolidated so that the resulting mining process may be more efficient, and the patterns found may be easier to understand.

Data Transformation Strategies overview:

1. <u>Smoothing</u>: Performed to remove noise.
   Ex: Binning, regression, clustering.
2. <u>Attribute construction</u>: New attributes are added to help mining process.
3. <u>Aggregation</u>: Data is summarized or aggregated.
   Ex: Sales data is aggregated into monthly & annual sales. This step is used for constructing data cube.
4. <u>Normalization</u>: Data is scaled so as to fall within a smaller range.
   Ex: -1.0 to +1.0.
5. <u>Data Discretization</u>: Where raw values are replaced by interval labels or conceptual labels.

Ex: Age

- Interval labels (10-18, 19-50)
- Conceptual labels (youth, adult)

6. Concept hierarchy generation for nominal data: Attributes are generalized to higher level concepts
   Ex: Street is generalized to city or country.

Data Transformation by Normalization:

The measurement unit used can affect data analysis. To help avoid dependence on the choice of measurement units, the data should be normalized or standardized. This involves transforming the data to fall within a smaller or common range such as Range = [-1,1], [0.0,1.0].

Normalizing the data attempts to give all attributes an equal weight For Ex: Changing unit from meters to inches in height lead to different results because of larger range for that attribute. To help avoid dependence on the choice of units, the data should be normalized.

Normalization attempts to give all attributes equal weight. Normalization is useful in classification algorithm involving neural networks or distance measurements such as nearest neighbor classification & clustering. There are different methods for normalization like - min-max normalization, z-score normalization, normalization by decimal scaling.

Min-Max Normalization:

a) Find min & max no. in the data.
b) Transform the data to range [new_{minA}, new_{maxA}] by computing

$$Vi' = \frac{Vi - minA}{maxA - minA}(new_{maxA} - new_{minA}) + new_{minA}$$

c) It preserves the relationship among the original data values
   Ex: If min income is Rs.12,000 & max income is Rs.98,000. If new range is [0.0,1.0].
   A value Vi= Rs.73600 will transform into

$$Vi' = \frac{73600 - 12000}{48000 - 12000}(1.0 - 0.0) + 0$$

Vi' = 0.716.

Z-score Normalization:

Values of an attribute A, are normalized on the mean & std deviation of A.

$$Vi' = \frac{Vi - A}{\sigma_A}$$ A= mean; σ_A= std deviation.

Also, variance(S_A) could be used which is more robust than std deviation(σ_A).

Decimal Scaling:

a) Normalizes by moving the decimal point of values.
b) The no. of decimal point moved depends on the max absolute value of A

$$Vi' = \frac{Vi}{10^j}$$

Ex: A= -986 to 917
Max Abs value = 986 (j=3)
Divide each no. by 1000 i.e. 10^3
Therefore -0.986 to 0.916 is the normalized value.